- Verify that euiccChallenge contained in serverSigned1 matches the one generated by the eUICC during step (3).
- Verify that euiccCiPKIdToBeUsed is supported and related credentials are available for signing.

If any verification fails, the eUICC SHALL return a relevant error status and the procedure SHALL be stopped.
If all the verifications succeed, the SM-XX is authenticated by the eUICC.

13. The eUICC SHALL:

    - Generate the euiccSigned1 data structure containing the TransactionID, serverChallenge, eUICCInfo2 and ctxParams1.
    - Compute the euiccSignature1 over euiccSigned1 using SK.EUICC.ECDSA. When generating the euiccSignature1, the eUICC SHALL use credentials related to the euiccCiPKIdToBeUsed parameter received from the SM-XX.

14. The eUICC SHALL return the euiccSigned1, euiccSignature1, CERT.EUICC.ECDSA, and CERT.EUM.ECDSA to the LPA.
15. The LPAd SHALL call the "ESXX.AuthenticateClient" function with input data including the euiccSigned1, euiccSignature1, CERT.EUICC.ECDSA, and the CERT.EUM.ECDSA.
16. On reception of the "ESXX.AuthenticateClient" function call, the SM-XX SHALL:

    - Correlate it with the "ESXX.InitiateAuthentication" function processed in step (7), by verifying the two TransactionIDs match.
    - Verify that the CERT.EUICC.ECDSA and CERT.EUM.ECDSA are valid as described in section 4.5.2.2.
    - Verify the euiccSignature1 performed over euiccSigned1 using the PK.EUICC.ECDSA contained in the CERT.EUICC.ECDSA.
    - Verify that serverChallenge contained in euiccSigned1 matches the one generated by the SM-XX during step (7).Verify that the SVN included in the euiccInfo2 is the same as the SVN which was delivered in the step (6).

    If any verification fails, the SM-XX SHALL return a relevant error status and the procedure SHALL be stopped.

17. This common call flow SHALL be followed by additional steps depending on the procedure within which it is used.

3.1.3 Profile Download and Installation

This section describes the Profile download and installation procedure.

```
@startuml
hide footbox
skinparam sequenceMessageAlign center
skinparam sequenceArrowFontSize 11
skinparam noteFontSize 11
skinparam monochrome true
skinparam lifelinestrategy solid

participant "<b>Operator" as OP
```

```
participant "<b>SM-DP+" as DP
participant "<b>LPAd" as LPA
participant "<b>eUICC" as E

rnote over LPA #FFFFFF : [1] (a) Get SM-DP+ Address, Parse Activation Code Token,
[SM-DP+ OID] from AC, or\n     (b) Get SM-DP+ Address and EventID from SM-DS, or\n
(c) Get Default SM-DP+ Address from eUICC

rnote over DP, E #FFFFFF : [2] [Refer to Common mutual authentication procedure
section 3.1.2]

rnote over DP #FFFFFF
[3]
- Look for Profile download pending order
- Eligibility Check using Device Info, euiccInfo2
endrnote

Group Opt.
DP -> OP : [4] ES2+.HandleDownloadProgressInfo(...)
OP --> DP : OK
end
DP --> LPA : [error]

rnote over DP #FFFFFF
[5]
- Build Profile Metadata
- Check if download retry
- Build smdpSigned2 = {TransactionID,
   Confirmation Code Required Flag, [bppEuiccOtpk]}
- Compute smdpSignature2 over smdpSigned2 and euiccSignature1
endrnote
DP -> LPA : [6] TransactionID, Profile Metadata, smdpSigned2, smdpSignature2,
CERT.DPpb.ECDSA

rnote over LPA #FFFFFF
[7a] Check if ProfileMetadata contains PPR(s)
endrnote

LPA -> E : [7b] [ES10b.GetRAT]
E --> LPA :  [RAT]

LPA -> E : [7c] [ES10b.GetProfilesInfo]
E --> LPA :  [ProfileInfoListOk]


rnote over LPA #FFFFFF
[8]
- [Check if PPR(s) is/are allowed against RAT.
Refer to section 2.9.2.3]
- [User Consent]
- [User Confirmation]
- [Prompt the End User to input Confirmation Code]
Endrnote

alt Download rejection
     rnote over DP, E #FFFFFF : [Refer to Sub-procedure Profile Download and
installation – Download rejection]

else Download confirmation
     rnote over DP, E #FFFFFF : [Refer to Sub-procedure Profile Download and
installation – Download confirmation]

end

@enduml
```

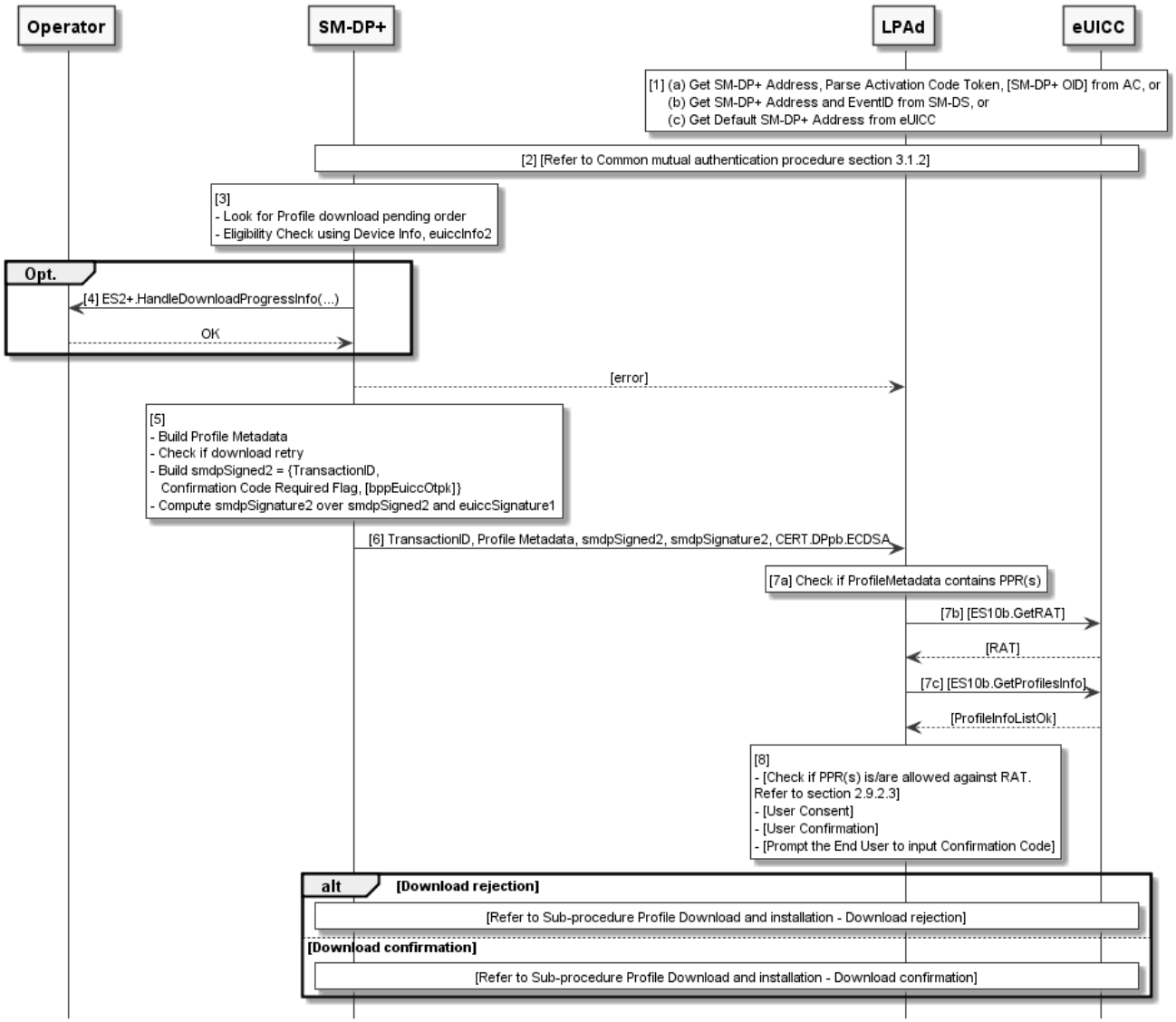

Figure 11: Profile Download and Installation

Start Conditions:

In addition to the start conditions required by the common mutual authentication procedure defined in section 3.1.2, this procedure requires the following start conditions depending on options in step 1:

- If this procedure uses an Activation Code (option a):
    - The End User has an Activation Code that is coded as described in the section 4.1.
    - The End User has entered the Activation Code to the LPAd. Depending on the Device capabilities, the LPAd SHALL support entry of the Activation Code by manual typing and QR code scanning.
- If this procedure uses an SM-DS (option b):

    - The LPAd has retrieved an SM-DP+ Address and EventID from the SM-DS.

- If this procedure uses a Default SM-DP+ (option c):
    - The LPAd has retrieved the Default SM-DP+ Address from the eUICC by calling the "ES10a.GetEuiccConfiguredAddresses" function.

Further, for each Profile the SM-DP+ SHALL maintain a count of the number of attempts to download that Profile and a count of the number of attempts to enter the Confirmation Code during download of that Profile. The SM-DP+ SHALL limit the number of download attempts and the number of Confirmation Code attempts, respectively.

A Provisioning Profile MAY be enabled by the LPAd upon End User request for RSP operations as defined in SGP.21 [4], which SHALL include End User consent if an Operational Profile is to be disabled and if establishment of the connectivity using the currently Enabled Profile is not successful.

Finally, if there is already an enabled Profile with PPR1 set, the following has occurred: The End User has been advised of this condition and has given consent for download. The LPA MAY alternatively request this consent at any later point during the download procedure.

Procedure:

1. (Optionally, i.e. for option (a)) The LPAd parses the Activation Code and finds the SM-DP+ address, Activation Code Token, and optional SM-DP+ OID. If the format of the Activation Code is invalid, the procedure SHALL be stopped with an error message provided by the LPAd to the End User.
2. The common mutual authentication procedure defined in section 3.1.2 SHALL be executed. When this procedure is used for Profile download and installation, SM-XX is SM-DP+. CERT.XXauth.ECDSA, PK.XXauth.ECDSA and SK.XXauth.ECDSA are CERT.DPauth.ECDSA, PK.DPauth.ECDSA and SK.DPauth.ECDSA respectively. ESXX is ES9+.

   During the common mutual authentication procedure at step (10), the LPAd SHALL verify that the SM-DP+ OID contained in the CERT.DPauth.ECDSA returned by the SM-DP+ is identical to the SM-DP+ OID if the LPAd has acquired it from the Activation Code at step (1). If the comparison fails, the LPAd SHALL inform the End User and the procedure SHALL be stopped.
   During the common mutual authentication procedure at step (10), the LPAd SHALL build the ctxParams1 data object to provide the MatchingID, Device Info to the eUICC for signature. The value of the MatchingID SHALL be set as follows:

   - If an Activation Code is used, the MatchingID value SHALL be set to Activation Code Token.
   - If an SM-DS is used, the MatchingID value SHALL be set to EventID.
   - If a Default SM-DP+ is used, the MatchingID SHALL be missing.

3. After having successfully authenticated the eUICC at the end of the step (2) above, the SM-DP+ SHALL:

- Verify that there is a related pending Profile download order for provided the MatchingID.
- If this Profile download order is already linked to an EID, verify that it matches the EID of the authenticated eUICC.
- Verify that the Profile corresponding to the pending Profile download order is in 'Released' state, or, in case of a retry due to a previous installation failure, in 'Downloaded' state (section 3.1.6).

If any of these verifications fail, the SM-DP+ SHALL return a relevant error status and the procedure SHALL be stopped.
The SM-DP+ SHALL increment the count of download attempts for the identified Profile. If the maximum number of attempts has been exceeded, the SM-DP+ SHALL terminate the corresponding Profile download order and notify the Operator by calling the "ES2+.HandleDownloadProgressInfo" function with an operation status indicating 'Failed' with the relevant error status, and the procedure SHALL be stopped.
Otherwise, the SM-DP+ SHALL perform appropriate eligibility checks, based on the Device Info and/or eUICCInfo2. These checks SHALL include the check if the eUICC can install one more Profile. See Annex F for more information on Eligibility checks.

4. (Optional step) Depending on the agreed behaviour with the Operator (out of scope of this specification), the SM-DP+ SHALL notify the Operator with the outcome of the eligibility check using the function "ES2+.HandleDownloadProgressInfo". The SM-DP+ SHALL provide the EID, the ICCID, the identification of the point reached (in that case it SHALL be 'Eligibility check'), the timestamp when this point was reached, and the execution result of this step.

   NOTE: This notification step MAY be done asynchronously.

5. If the eligibility check fails, the SM-DP+ SHALL:

   - Set the Profile corresponding with the pending Profile download order in 'Error' state (section 3.1.6).

   - Return an error status to the LPAd and the procedure SHALL be stopped.

   Otherwise, the SM-DP+ SHALL:

   - Determine whether the Profile is already bound to the EID from a previous unsuccessful download attempt. If so, the SMDP+ MAY include the otPK.eUICC.ECKA obtained in the previous session in the smdpSigned2 data structure.
   - Determine if a Confirmation Code is required for this pending order.
   - Generate a smdpSigned2 data structure containing the TransactionID and the Confirmation Code Required Flag.
   - Compute the smdpSignature2 over smdpSigned2 and euiccSignature1 using the SK.DPpb.ECDSA.

6. The SM-DP+ returns the TransactionID, ProfileMetadata, smdpSigned2, smdpSignature2 and CERT.DPpb.ECDSA to the LPAd.

7. On reception of the SM-DP+ response, the LPAd SHALL check if the ProfileMetadata contains PPR(s).

a) If the ProfileMetadata contains PPR(s), the target eUICC is removable and the LPAd does not support download of Profiles with PPRs regardless of the RAT in the removable eUICC, then the LPAd SHALL continue the Sub-procedure "Profile Download and installation – Download rejection" hereunder with reason code ‘PPR not allowed’.

b) If the ProfileMetadata contains PPR(s) and the LPAd does not already have the Rules Authorisation Table, then the LPAd SHALL request the Rules Authorisation Table from the eUICC by calling the "ES10b.GetRAT" function.

c) If the ProfileMetadata contains PPR1 and the LPAd does not already have the list of installed Profiles, then the LPAd SHALL request the information from the eUICC by calling the "ES10b.GetProfilesInfo" function. If the ProfileMetadata contains PPR1 and an Operational Profile is installed, the LPAd SHALL perform the Sub-procedure "Profile Download and installation - Download rejection" hereunder with reason code 'PPR not allowed'.

8. If the ProfileMetadata contains PPR(s), the LPAd SHALL check if the PPR(s) is/are allowed based on the Rules Authorisation Table defined in section 2.9.2.3. If one or more PPR(s) are not allowed, the LPAd SHALL continue the Sub-procedure "Profile Download and installation – Download rejection" hereunder with reason code ‘PPR not allowed’. If any PPR is subject to additional End User consent according to the RAT, LPAd SHOULD ask for Strong Confirmation by showing relevant information concerning the PPR(s). This information SHOULD include the consequences of the Profile Policy Rule to the End User. This message SHALL be formulated in a descriptive and non-discriminatory manner (e.g. for "Non-Delete" Profile Policy Rule: "The profile that you are about to install can be deleted only under the terms you have agreed with your service provider. Enter your PIN to approve installation"). If the Profile Metadata does not contain any Profile Policy Rule(s) subject to additional End User consent, the LPAd SHALL ask for Simple Confirmation (e.g., simple 'Yes' or 'No' or 'Not Now') on the Profile download.

   If the End User does not agree to the Profile Policy Rules, the LPAd SHALL continue the Sub-procedure "Profile Download and installation – Download rejection" hereunder with reason code 'End User rejection' or 'postponed'.
   If the Confirmation Code Required Flag is set in either the Activation Code Token or in the smdpSigned2, then the LPAd SHALL ask for the End User to enter the Confirmation Code which was provided by the Operator that MAY be considered as a Strong Confirmation. When prompting, the LPAd MAY also display the ProfileName or any relevant information contained in the Profile Metadata to help the End User identify the Profile to be downloaded during this RSP session. If the Confirmation Code is not required, the LPAd SHALL ask for Simple Confirmation (e.g. simple 'Yes' or 'No' or 'Not Now') by showing the ProfileName or any relevant information contained in the Profile Metadata to the End User either at this step or after the BPP has been downloaded by the LPAd, since the same Profile Metadata will also be available then.

   The Confirmation Requests in this step MAY be combined into a single prompt therefore requiring a single confirmation by the End User. Whether combined or

separated, these can be performed either at this step or after the BPP has been downloaded by the LPAd, since the same Profile Metadata will also be available then. However, this exception does not apply to the Confirmation Code input which SHALL be obtained no later than this step.

The case where the End User does not agree to the download of the Profile (e.g. by selecting 'No' or 'Not Now'), is described in the Sub-procedure "Profile Download and Installation – Download Rejection" hereunder.

If the End User does not respond to the LPAd prompt within an implementation-dependent timeout interval, the LPAd SHALL cancel the Profile download by performing the sub-procedure "Profile Download and Installation – Download Rejection" hereunder with the reason 'Timeout'.

If required, the LPAd SHALL calculate the hash of the Confirmation Code as follows:

Hashed Confirmation Code = SHA256 (SHA256 (Confirmation Code) | TransactionID), where '|' means concatenation of data.

If Profile download has not been rejected in the steps above, the procedure SHALL continue with the Sub-procedure "Profile Download and installation – Download confirmation"

3.1.3.1 Sub-procedure Profile Download and Installation – Download Rejection

This procedure can occur due to an End User rejection or timeout of a Profile download at the following steps of the protocol:

- after the response to "ES9+.Authenticate Client" (section 3.1.3), and
- after the response to "ES9+.GetBoundProfilePackage" (section 3.1.3.2).

The LPAd MAY provide additional places where the End User would be offered such possibility.

This procedure can also occur if the LPAd detects a mismatch in some of the Profile Metadata returned by "ES9+.Authenticate Client" and "ES9+.GetBoundProfilePackage".

```
@startuml
hide footbox
skinparam sequenceMessageAlign center
skinparam sequenceArrowFontSize 11
skinparam noteFontSize 11
skinparam monochrome true
skinparam lifelinestrategy solid

participant "<b>Operator" as OP
participant "<b>SM-DP+" as DP
participant "<b> LPAd" as LPA
participant "<b>eUICC" as E

LPA -> E : [1] ES10b.CancelSession(TransactionID, reason)
rnote over E #FFFFFF
[2]
- Generate euiccCancelSessionSigned = {
   TransactionID, reason}
- Compute euiccCancelSessionSignature
   over euiccCancelSessionSigned
endrnote
```

```
E --> LPA : [3] euiccCancelSessionSigned, \neuiccCancelSessionSignature
LPA -> DP : [4] ES9+.CancelSession(TransactionID, \neuiccCancelSessionSigned,
euiccCancelSessionSignature)

rnote over DP #FFFFFF
[5]
- Verify euiccCancelSessionSignature
- Get the reason
endrnote
DP --> LPA : [error]

Group Cond. Download rejection
rnote over DP #FFFFFF
[6]
- Terminate Download order
- [Delete Event, Refer to Event Deletion section 3.6.3]
endrnote

DP -> OP : [7] ES2+.HandleDownloadProgressInfo
OP --> DP : OK

end

DP --> LPA : OK

@enduml
```

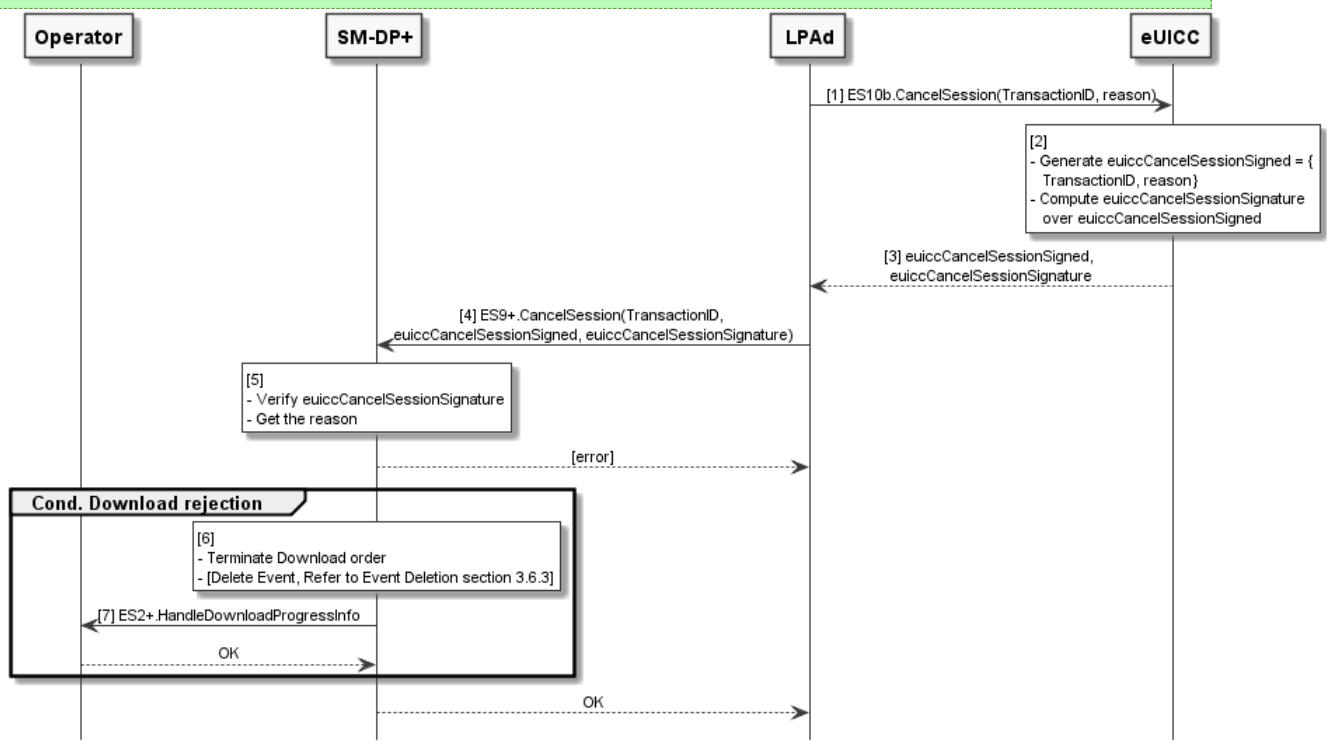

Figure 12: Sub-procedure Profile Download and Installation - Download Rejection

Start Conditions:

The End User has not agreed to the download of the Profile (e.g. by selecting 'No'), the PPR(s) in the ProfileMetadata is/are not allowed according to the Rules Authorisation Table, PPR1 is present in the ProfileMetadata and an Operational Profile is already installed on the eUICC, the download of the Profile containing PPR(s) onto a removable eUICC (regardless of its RAT) is not supported by the LPAd, the ProfileMetadata in the response to

"ES9+.Authenticate Client" does not match the ProfileMetadata in the Bound Profile Package, or the LPAd has encountered an error while installing the Bound Profile Package.

Procedure:

1. The LPAd SHALL call the "ES10b.CancelSession" function with input data including the TransactionID and the reason 'End User rejection', 'End User postponed' or 'Timeout', ‘PPR not allowed’, 'Metadata mismatch', 'Load BPP execution error' or 'undefinedReason'.
2. On reception of this function call, the eUICC SHALL:
   - Generate the euiccCancelSessionSigned data object containing the TransactionID and the reason provided by the LPAd.
   - Compute the euiccCancelSessionSignature over euiccCancelSessionSigned using the SK.EUICC.ECDSA corresponding to the euiccCiPKIdToBeUsed as received during the Common mutual authentication procedure.
3. The eUICC SHALL return the euiccCancelSessionSigned and euiccCancelSessionSignature.
4. The LPAd SHALL call the "ES9+.CancelSession" function with input data including the TransactionID, the euiccCancelSessionSigned and the euiccCancelSessionSignature.
5. On reception of the "ES9+.CancelSession" function, the SM-DP+ SHALL:
   - Retrieve the on-going RSP session identified by the TransactionID. If the TransactionID is unknown, the SM-DP+ SHALL return a function execution status 'Failed' with relevant status code. The LPAd MAY retry this step with a different TransactionID if the provided value was incorrect.
   - Verify the euiccCancelSessionSignature performed over euiccCancelSessionSigned using the PK.EUICC.ECDSA associated with the ongoing RSP session. If the signature is invalid, the SM-DP+ SHALL return a function execution status ‘Failed’ with relevant status code and the procedure SHALL be stopped by the LPAd.
   - Verify that the received smdpOid corresponds to the SM-DP+ (i.e. is the same value as the one contained in the CERT.DPauth.ECDSA used during the Common Mutual Authentication Procedure). If the value doesn't match, the SM-DP+ SHALL return a function execution status 'Failed' with relevant status code and the procedure SHALL be stopped by the LPAd.

   If the reason contained in euiccCancelSessionSigned indicates 'End user postponed' or 'Timeout', the SM-DP+ SHALL simply return a function execution status 'Executed-Success' and keep the corresponding Profile download order in the 'Released' state available for a further retry, and the procedure SHALL be stopped.
   If the reason contained in euiccCancelSessionSigned indicates any other condition the SM-DP+ SHALL perform the following steps.
6. The SM-DP+ SHALL set the Profile associated with the on-going RSP session in 'Error' state (section 3.1.6); and if this procedure is executed in the context of option (b), the SM-DP+ SHALL execute the SM-DS event deletion procedure described in section 3.6.3.

7. The SM-DP+ SHALL call the "ES2+.HandleDownloadProgressInfo" function with the relevant notificationPointId set and an operation status indicating 'Failed' with status code value depending on the given cancel reason. The cancel session reason code mapping to status code is given in section 5.3.5.

   The SM-DP+ SHALL return a function execution status 'Executed-Success' and the procedure SHALL be stopped.

3.1.3.2 Sub-procedure Profile Download and Installation – Download Confirmation

```
@startuml
hide footbox
skinparam sequenceMessageAlign center
skinparam sequenceArrowFontSize 11
skinparam noteFontSize 11
skinparam monochrome true
skinparam lifelinestrategy solid

participant "<b>Operator" as OP
participant "<b>SM-DP+" as DP
participant "<b> LPAd" as LPA
participant "<b>eUICC" as E


LPA -> E : [1] ES10b.PrepareDownload \n (smdpSigned2, smdpSignature2,
CERT.DPpb.ECDSA, [Hashed Confirmation Code])
rnote over E #FFFFFF
[2]
- Verify CERT.DPpb.ECDSA
- Verify CERT.DPauth.ECDSA and CERT.DPpb.ECDSA have same owner
- Verify smdpSignature2 over smdpSigned2
- Verify smdpSigned2
Endrnote
E --> LPA: [error]
rnote over E #FFFFFF
[3]
- Generate one time ECKA key pair
   (otPK.EUICC.ECKA, otSK.EUICC.ECKA)
unless a valid otPK.EUICC.ECKA was provided
- Generate euiccSigned2=
   {TransactionID, otPK.EUICC.ECKA, [Hashed Confirmation Code]}
- Compute euiccSignature2 over euiccSigned2 and smdpSignature2
Endrnote
E -> LPA: [4] euiccSigned2, euiccSignature2
LPA -> DP : [5] ES9+.GetBoundProfilePackage \n (euiccSigned2, euiccSignature2)

rnote over DP #FFFFFF
[6]
- Verify euiccSignature2 over euiccSigned2
- Determine if Confirmation Code required
Endrnote
DP --> LPA : [error]

Group Cond. Confirmation Code handling
rnote over DP #FFFFFF
[7]
- [Verify Hashed Confirmation Code]
Endrnote
Group Cond. On CC error
DP -> OP : [8] ES2+.HandleDownloadProgressInfo
OP --> DP : OK
DP --> LPA : [error]
end
end
```

```
rnote over DP #FFFFFF
[9]
- [Generate one time ECKA key pair (otPK.DP.ECKA, otSK.DP.ECKA)]
- [Generate Session Keys]
- Generate Bound Profile Package
Endrnote

Group Opt.
DP -> OP : [10] ES2+.HandleDownloadProgressInfo(...)
OP --> DP : OK
end

DP --> LPA : [11] TransactionID, Bound Profile Package

rnote over LPA #FFFFFF
[12]
- [Verify Metadata]
- [Prompt/Display Profile Metadata to End User]
Endnote

alt
     rnote over OP, E #FFFFFF : [13] [Refer to Sub-procedure Profile Download and
installation – Download rejection]
else
     rnote over OP, E #FFFFFF : [14] [Refer to Sub-procedure Profile Installation]
end
@enduml
```

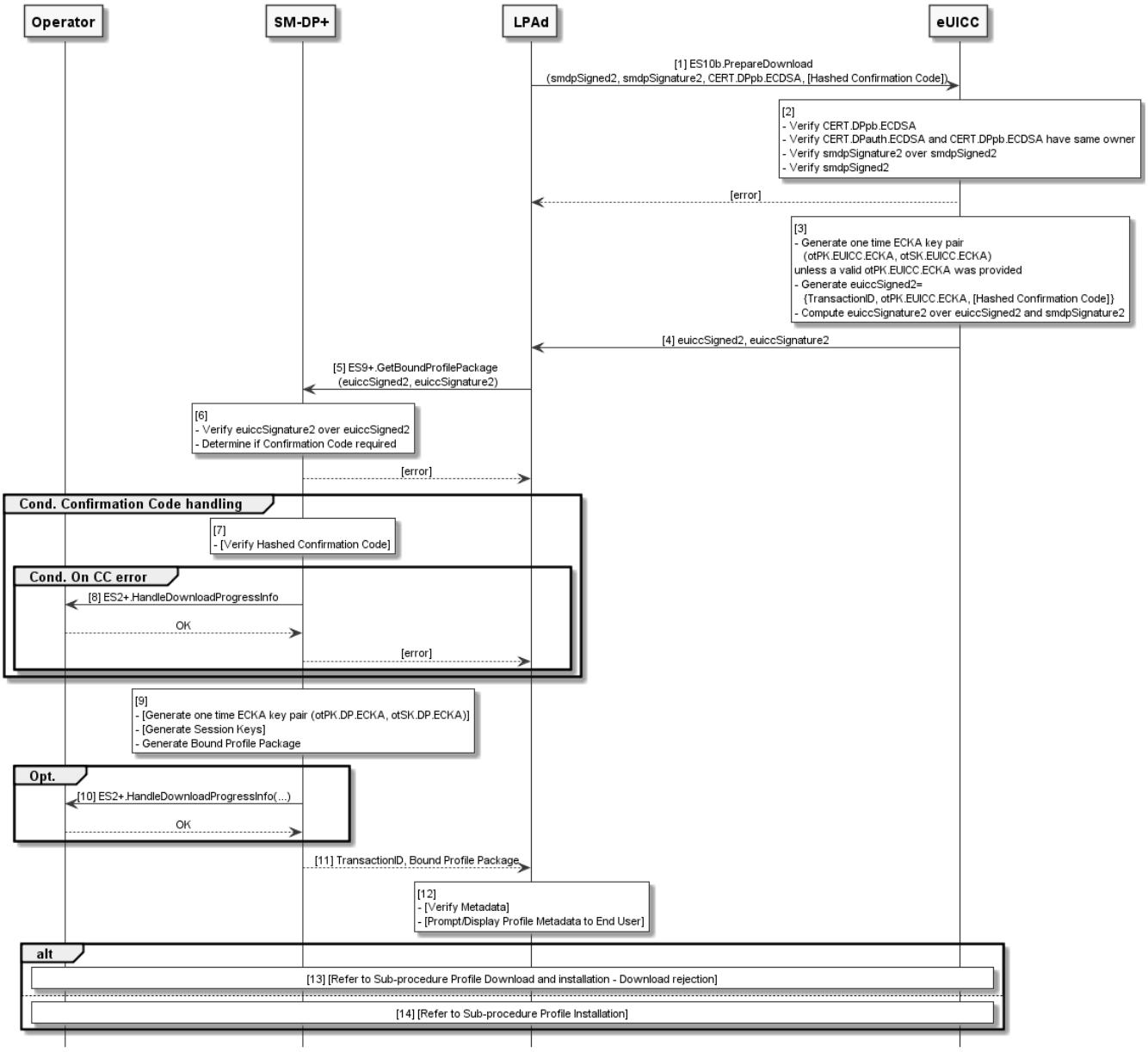

Figure 13: Sub-procedure Profile Download and Installation – Download Confirmation

Start Conditions:

The End User has agreed to the download of the Profile (e.g. by selecting 'Yes').

Procedure:

1. The LPAd SHALL call the "ES10b.PrepareDownload" function with input data including the smdpSigned2, smdpSignature2, CERT.DPpb.ECDSA and optionally the Hashed Confirmation Code calculated during the last step of the profile download and installation procedure (section 3.1.3).
2. On reception of the "ES10b.PrepareDownload" function, the eUICC SHALL:

   - Verify that CERT.DPpb.ECDSA is valid according to section 4.5.2.2.
   - Verify that CERT.DPauth.ECDSA and CERT.DPpb.ECDSA belong to the same entity (i.e. same OID in, same subjectAltName).

- Verify smdpSignature2 performed over smdpSigned2 and euiccSignature1 using the PK.DPpb.ECDSA contained in CERT.DPpb.ECDSA.
- Verify that the TransactionID contained in smdpSigned2 matches the TransactionID of the on-going RSP session.
- Verify that the Hashed Confirmation Code is provided by the LPAd if the Confirmation Code Required Flag is set in smdpSigned2.

If any of the verifications fail, the eUICC SHALL return a relevant error status and the procedure SHALL be stopped.

3. Otherwise the eUICC SHALL:

   - If bppEuiccOtpk is provided in smdpSigned2 and it corresponds to a stored one-time key pair (otPK.EUICC.ECKA, otSK.EUICC.ECKA) for this SM-DP+, the eUICC SHALL use this key pair for the RSP session. Otherwise it SHALL generate a new one-time key pair (otPK.EUICC.ECKA, otSK.EUICC.ECKA) using the curve indicated by the Key Parameter Reference Value of CERT.DPpb.ECDSA.
   - Generate the euiccSigned2 data structure containing the TransactionID, otPK.EUICC.ECKA and optionally the Hashed Confirmation Code.
   - Compute the euiccSignature2 over euiccSigned2 and smdpSignature2 using SK.EUICC.ECDSA. When generating the euiccSignature2, the eUICC SHALL use the same private key as in the AuthenticateServer response.

4. The eUICC SHALL return the euiccSigned2 and euiccSignature2.
5. The LPAd calls the "ES9+.GetBoundProfilePackage" function with input data including the euiccSigned2, euiccSignature2.
6. On reception of the "ES9+.GetBoundProfilePackage" function, the SM-DP+ SHALL verify the euiccSignature2 performed over euiccSigned2 and smdpsignature2 using the PK.EUICC.ECDSA associated with the RSP session identified by TransactionID.
   If a Confirmation Code is required the SM-DP+ SHALL continue with step 7. Otherwise, the SM-DP+ SHALL continue at step 9.
7. If a Confirmation Code verification is required, the SM-DP+ SHALL:

   - Retrieve the hashed Confirmation Code stored for this order by "ES2.ConfirmOrder" and calculate the expected hash value as

     expected hash value =
     SHA256(stored hashed Confirmation Code | TransactionID)

   - Verify that the received Hashed Confirmation Code matches the expected hash value.
   - In case the Confirmation Code verification has failed, the SM-DP+ SHALL increment the count of Confirmation Code attempts for the Profile. If the maximum number of retries has been exceeded, the SM-DP+ SHALL set the Profile corresponding to the Profile download order in 'Error' state (section 3.1.6) and execute step 8.

8. (Conditional) The SM-DP+ SHALL notify the Operator by calling "ES2+.HandleDownloadProgressInfo" with the corresponding identification point

reached (in that case it SHALL be 'Confirmation Code check') and the operation status (Execution status = 'Failed' with the relevant status code value).

If any verification in step 6 or 7 failed, the SM-DP+ SHALL return an error status to the LPAd and the procedure SHALL be stopped. Unless the maximum number of retries for Confirmation Code entries has been exceeded, the LPAd MAY retry by restarting the Profile download and installation procedure.

9. Otherwise, the SM-DP+ SHALL perform the following:

   - If a re-usable BPP exists and the eUICC returned the corresponding otPK.EUICC.ECKA, only the signature for InitialiseSecureChannel needs to be recalculated and the next steps can be skipped.
   - If the re-usable BPP exists and the eUICC returned a different otPK.EUICC.ECKA, the SM-DP+ SHALL either return an error to the LPAd and terminate the procedure, or re-bind the PPP as described in section 5.6.2.
   - If the Profile is not linked to the EID of the target eUICC, the SM-DP+ SHALL link the Profile to the EID at this step.
   - Generate a one-time ECKA key pair (otPK.DP.ECKA, otSK.DP.ECKA) using the curve indicated by the Key Parameter Reference Value of CERT.DPpb.ECDSA.
   - Generate Session Keys using the CRT, otPK.eUCC.ECKA, and otSK.DP.ECKA according to Annex G.
   - Prepare the Bound Profile Package according to the description given in section 2.5.4, including optionally the Profile Protection Keys (PPK).

10. (Optional step) Depending on the agreed behaviour with the Operator (out of scope of this specification), the SM-DP+ SHALL notify the Operator that the Profile is downloaded using the function "ES2+.HandleDownloadProgressInfo". The SM-DP+ SHALL provide the EID, the ICCID, the identification of the point reached (in that case it SHALL be 'BPP download'), the timestamp when this point was reached, and the execution result of this step.

    NOTE: This notification step MAY be done asynchronously.

11. The SM-DP+ responds back to the LPAd with the TransactionID and the Bound Profile Package and set the Profile corresponding to the Profile download order in 'Downloaded' state (section 3.1.6).
12. On reception of the SM-DP+ response, the LPAd MAY perform additional processing using the Profile Metadata contained within the Bound Profile Package:

    - If the LPAd previously used the Profile Metadata returned by "ES9+.AuthenticateClient" (i.e., in step (7 and 8) of the procedure described in section 3.1.3), then

      - The LPAd SHALL verify that the Profile Policy Rules have not changed. If this verification fails, the LPAd SHALL execute the Sub-procedure "Profile Download and installation – Download Rejection" (section 3.1.3.1) with reason code 'PPR not allowed'.
      - The LPAd SHOULD verify that all other Profile Metadata elements it used in that earlier step (such as the Profile Name, Icon, etc.) have not changed. If the

verification fails, the LPAd MAY inform the End User and offer the End User to postpone or reject the Profile installation. Alternatively, the LPAd MAY stop the download by executing the Sub-procedure "Profile Download and installation – Download Rejection" (section 3.1.3.1) with reason code 'Metadata mismatch'.

- If the LPA has not previously captured the End User consent related to Profile Metadata and any Profile Policy Rules, as defined in section 3.1.3 step (8), it SHALL do so at this point.
- The LPAd MAY display any relevant part of the Profile Metadata to help the End User identify the Profile to be installed during this transaction. Based on this information, the LPAd MAY offer the End User to postpone or reject the Profile installation. If the End User does not respond to the LPAd prompt within an implementation-dependent timeout interval, the LPAd SHALL cancel the Profile download by performing the sub-procedure "Profile Download and Installation – Download Rejection" described in section 3.1.3.1 with the reason 'Timeout'.

13. If End User has postponed or rejected the Profile installation, the Sub-procedure "Profile Download and Installation – Download Rejection" described in section 3.1.3.1 SHALL be executed.
14. Otherwise sub-procedure Profile installation described in section hereafter SHALL be executed.

3.1.3.3 Sub-procedure Profile Installation

```
@startuml
hide footbox
skinparam sequenceMessageAlign center
skinparam sequenceArrowFontSize 11
skinparam noteFontSize 11
skinparam monochrome true
skinparam lifelinestrategy solid

participant "<b>Operator" as OP
participant "<b>SM-DP+" as DP
participant "<b> LPAd" as LPA
participant "<b>eUICC" as E

LPA -> E : [1] ES10b.LoadBoundProfilePackage x N\n(ES8+.InitialiseSecureChannel)

rnote over E #FFFFFF
[2]
- Verify InitialiseSecureChannel data
- Generate Session Keys
endnote
E --> LPA : Response APDU x N
LPA -> E : [3] ES10b.LoadBoundProfilePackage x N\n(ES8+.ConfigureISDP)
E --> LPA : Response APDU x N
LPA -> E : [4] ES10b.LoadBoundProfilePackage x N\n(ES8+.StoreMetadata)
E --> LPA : Response APDU x N

Group Cond. ES8+.StoreMetadata contains PPR(s)
rnote over E #FFFFFF
[4a] Verify PPR(s) against RAT. Refer to section 2.9.3.1
endrnote
end

LPA -> E : [5] [ES10b.LoadBoundProfilePackage x N]\n(ES8+.ReplaceSessionKeys)
E --> LPA : [Response APDU x N]
LPA -> E : [6] ES10b.LoadBoundProfilePackage x N\n(ES8+.LoadProfileElements)
E --> LPA : Response APDU x N \n(ProfileInstallationResult)
```

```
LPA -> DP : [7] ES9+.HandleNotification(ProfileInstallationResult)
DP --> LPA : OK

rnote over DP #FFFFFF
[8] [Terminate Download order]
endrnote

DP -> OP : [9] [ES2+.HandleDownloadProgressInfo]
OP --> DP : OK

rnote over DP #FFFFFF
[10] [Delete Event, Refer to Event Deletion section 3.6.3]
endrnote

LPA -> E : [11] ES10b.RemoveNotificationFromList

rnote over E #FFFFFF
[12] Delete Notification
endrnote

E --> LPA : OK

@enduml
```

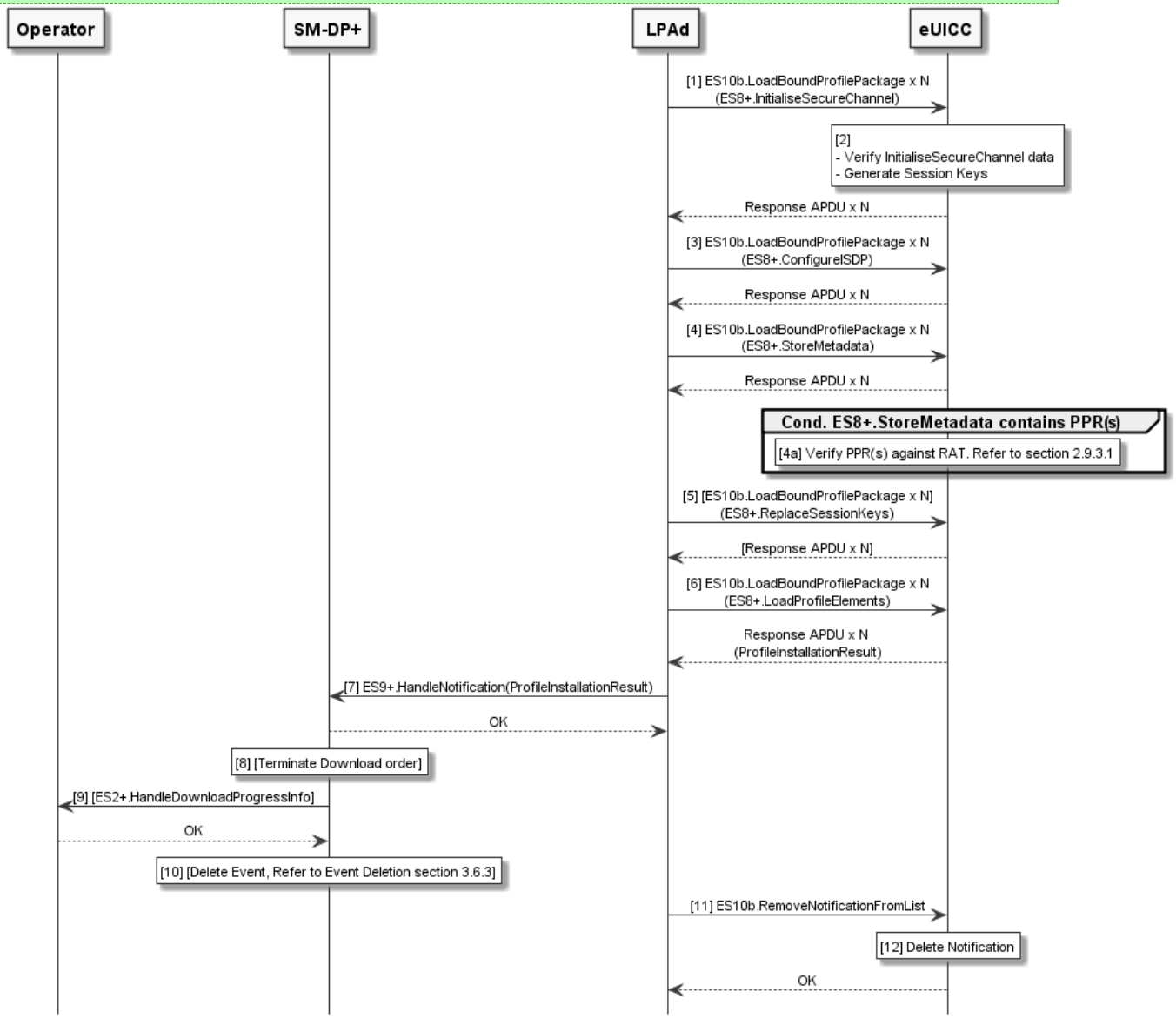

Figure 14: Sub-procedure Profile Installation

In this sub-procedure the LPAd generates the Segmented Bound Profile Package according to the description in section 2.5.5 and transfers it to the eUICC using a sequence of "ES10b.LoadBoundProfilePackage" commands. If the LPAd is unable to perform the segmentation (e.g., because of an error in the BPP structure), or if any call of "ES10b.LoadBoundProfilePackage" returns status words other than '90 00' or '91 XX', the LPAd SHALL perform the Sub-procedure "Profile Download and installation – Download rejection" with reason code 'Load BPP execution error'.

1. The LPAd SHALL transfer the "ES8+.InitialiseSecureChannel" function call included in the Bound Profile Package to the eUICC by repeatedly calling the "ES10b.LoadBoundProfilePackage" function. The input data of the "ES8+.InitialiseSecureChannel" function includes the CRT, otPK.DP.ECKA and smdpSign2.
2. The eUICC SHALL verify the received "ES8+.InitialiseSecureChannel" according to the description in section 5.5.1. If the verification succeeds, the eUICC SHALL generate Session Keys using the input data received in previous step according to Annex G.
3. The LPAd SHALL transfer the "ES8+.ConfigureISDP" function call included in the Bound Profile Package to the eUICC by repeatedly calling the "ES10b.LoadBoundProfilePackage" function.
4. The LPAd SHALL transfer the "ES8+.StoreMetadata" function call included in the Bound Profile Package to the eUICC by repeatedly calling the "ES10b.LoadBoundProfilePackage" function.
4. a) If "ES8+.StoreMetadata" contains PPR(s), the eUICC SHALL verify each PPR according to PPR verification section 2.9.3.1.
5. If the Profile Protection Keys (PPK) were included in the Bound Profile Package, the LPAd SHALL transfer the "ES8+.ReplaceSessionKeys" function call to the eUICC by repeatedly calling the "ES10b.LoadBoundProfilePackage" function. The input data of the "ES8+.ReplaceSessionKeys" function includes the PPK. Once the eUICC receives "ES8+.ReplaceSessionKeys" function call, it SHALL decrypt the Profile Protection Keys and replace the current SCP03t Session Keys with the decrypted Profile Protection Keys.
6. The LPAd SHALL transfer the Profile Elements included in the "ES8+.LoadProfileElements" functions by repeatedly calling the "ES10b.LoadBoundProfilePackage" function.

   If all the Profile Elements are successfully processed and installed, with or without any warning, the last response of the "ES10b.LoadBoundProfilePackage" function SHALL deliver the Profile Installation Result, comprising the `ProfileInstallationResultData` as defined in section 2.5.6 and the `EuiccSignPIR` which is the signature generated across the `ProfileInstallationResultData` data object using SK.EUICC.ECDSA). Installation notifications as configured in StoreMetadata, if any, SHALL be generated.

   Otherwise, if an error occurs during the transfer and processing of the Profile Elements, or any previous ES8+ function, which does not result in error status words, the eUICC SHALL stop the procedure and SHALL report to the LPAd with a response of the "ES10b.LoadBoundProfilePackage" function including the Profile Installation Result.

The eUICC SHALL store the Profile Installation Result in its non-volatile memory before delivering it to the LPAd.
The eUICC SHALL erase the otSK.EUICC.ECKA attached to this RSP session no later than the successful completion of the BPP installation.

7. The LPAd calls the "ES9+.HandleNotification" function with input data including the Profile Installation Result which was given by the eUICC in the step 6.
8. On reception of the "ES9+.HandleNotification" function, the SM-DP+ SHALL:
   - Acknowledge the reception of the notification to the LPAd.
   - Retrieve the pending download order identified by the TransactionID. If TransactionID is unknown, the SM-DP+ SHALL terminate its processing.
   - (Conditional) Terminate the pending download order and set the corresponding Profile in state 'Installed' or 'Error' (section 3.1.6) as indicated by the Profile Installation Result.
9. (Conditional) The SM-DP+ SHALL call the "ES2+.HandleDownloadProgressInfo" with input data including eid, iccid, ProfileType, timestamp, resultData, identification of the point reached (in that case it SHALL be 'BPP installation') and notificationPointStatus set accordingly to Profile Installation Result.
10. (Conditional) If this procedure is executed in the context of option (b), the SM-DP+ SHALL execute the SM-DS event deletion procedure (section 3.6.3).
11. On reception of the acknowledgement message from the SM-DP+ the LPAd SHALL call "ES10b.RemoveNotificationFromList" with corresponding seqNumber as input parameter.
12. The eUICC SHALL delete the Profile Installation Result from its non-volatile memory.

3.1.4 Limitation for Profile Installation

Several profiles MAY be installed on the eUICC, subject to non-volatile memory limitations.

3.1.5 Error Handling Within the Profile Download Procedure

The Profile download and installation procedure comprises a sequence of operations between the SM-DP+, the LPA, and the eUICC over a period of time. In addition to errors reported by ES9+ and ES10b functions, other conditions MAY impact the successful execution of this procedure. The LPA SHOULD indicate such failures to the user; however, the specific presentation of these errors is out of the scope of this document.

The LPA SHOULD NOT initiate a new Profile download and installation procedure while there is an active download RSP session. However, in the event that this does occur, the eUICC SHALL discard its session state (including generated eUICC challenge, any downloaded Profile Metadata, Profile contents, and Profile Installation Result) with the possible exception that an unused otPK/otSK.EUICC.ECKA MAY be stored for future retry, when a new RSP session is started with "ES10b.GetEUICCChallenge".

The eUICC MAY discard its session state if a Profile switch occurs during a Profile download and installation procedure.

If an eUICC Memory Reset or eUICC Test Memory Reset is successfully processed during a Profile download and installation procedure, the eUICC SHALL discard its session state.

If the eUICC receives a BPP segment with an unrecognized leading tag (see section 2.5.5) during Profile download it SHALL return status words of '6A 88' (Reference data not found) and SHALL not discard the download session state.

The Profile download and installation procedure MAY fail because of a communications failure between the LPA and the SM-DP+. The LPA MAY retry for a period of time. The LPA SHALL reset its own Profile download session state when all retry attempts have failed.

While the SBPP is sent to the eUICC using "ES10b.LoadBoundProfilePackage", the eUICC MAY reject any other ES10 command with status words '69 85' (Conditions of use not satisfied), except "ES10b.GetEUICCChallenge" (indicating the start of a new download session) or "ES10b.CancelSession" (indicating the termination of the current download session).

The Profile download and installation procedure could fail while the LPA is sending SBPP TLVs to the eUICC using "ES10b.LoadBoundProfilePackage" for reasons other than an error status reported by the eUICC. Examples of such failures during the download process include:

- In the case of a removable eUICC card, the End User could remove the card.
- The End User could switch off the power or remove the battery.
- A software fault could cause a crash of the LPA, host Device, and/or baseband processor.

The LPA SHOULD provide an appropriate error indication to the End User when possible (e.g., when power is restored). The specific presentation of such an error notification is out of scope of this document.

3.1.6 Profile Lifecycle at SM-DP+

The previous sections provide detailed procedures associated with Remote Provisioning. Each Profile has state information on the SM-DP+ associated with it during the provisioning into an eUICC. The Profile lifecycle state can be one of the states listed in the following table.

Additional states and additional or customised ES2+ functions MAY be agreed between the Operator and the SM-DP+.

State Name	Description
Available	The Profile is available in the inventory of the SM-DP+.
Allocated	The Profile is reserved for downloading without being linked to an EID.
Linked	The Profile is reserved for downloading and is linked to an EID.
Confirmed	The Profile is reserved for downloading (linked or not linked to an EID) with Matching ID and Confirmation Code if required.
Released	The Profile is ready for download and installation after Network Configuration by the Operator (e.g.: HLR Registration).
Downloaded	The Bound Profile was delivered to the LPA.
Installed	The Profile was successfully installed on the eUICC.